

# Tier 5: Subgroup Fairness Analysis

Abdul Ahad. Student ID: G00486649

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### 0.1 Overview

Subgroup discrimination and calibration of the Tier-1 nomogram's out-of-sample risk are assessed here, stratified by sex and age band. For each subgroup the sample size, event rate, AUROC, and calibration-in-the-large (mean predicted risk versus observed rate) are reported. External validation on eICU is deferred (no local database available).

### 0.2 Load Libraries

```
suppressPackageStartupMessages({ library(pROC) })
```

### 0.3 Locate and Read Data

```
find_data <- function(f) {  
  cand_s <- c(file.path("../processed_data", f),  
              file.path("processed_data", f), f)  
  hit <- cand_s[file.exists(cand_s)][1]; if (is.na(hit)) stop("cannot find ", f); hit  
}  
OUT <- dirname(find_data("10_tier1_oos_pred.parquet"))  
  
pred <- as.data.frame(arrow::read_parquet(find_data("10_tier1_oos_pred.parquet")))  
lm6 <- as.data.frame(arrow::read_parquet(find_data("08_landmark_h6.parquet")))[, c("stay_id", "a  
d <- merge(pred, lm6, by="stay_id")  
d$age_band <- cut(d$age, c(-Inf, 50, 65, 80, Inf),  
                 labels=c("<50", "50-64", "65-79", "80+"))
```

### 0.4 Define Subgroup Metrics Function

For each subgroup, AUROC, mean predicted risk, observed event rate, and their ratio (calibration-in-the-large) are computed. A calibration ratio of 1.0 indicates that the average prediction matches the

true event rate; values above 1 indicate over-prediction, values below 1 under-prediction.

```
grp_metrics <- function(sub, label) {
  y <- sub$onset_within_h; p <- sub$risk_en
  au <- tryCatch(as.numeric(auc(roc(y, p, quiet=TRUE, direction="<"))), error=function(e) NA)
  data.frame(subgroup=label, n=nrow(sub), event_rate=round(mean(y),4),
             auroc=round(au,4), mean_pred=round(mean(p),4),
             obs_rate=round(mean(y),4),
             calib_ratio=round(mean(p)/mean(y),3))
}
```

## 0.5 Compute Subgroup Metrics and Write Results

Fairness metrics are computed for the full cohort and for each sex and age-band subgroup, then saved to tier5\_subgroups.csv.

```
rows <- list(grp_metrics(d, "ALL"))
for (g in sort(unique(d$gender)))
  rows[[length(rows)+1]] <- grp_metrics(d[d$gender==g,], paste0("sex=",g))
for (b in levels(d$age_band))
  rows[[length(rows)+1]] <- grp_metrics(d[d$age_band==b,], paste0("age=",b))
tab <- do.call(rbind, rows)
print(tab, row.names=FALSE)
```

| ## | subgroup  | n     | event_rate | auroc  | mean_pred | obs_rate | calib_ratio |
|----|-----------|-------|------------|--------|-----------|----------|-------------|
| ## | ALL       | 48829 | 0.0682     | 0.7752 | 0.0682    | 0.0682   | 1.000       |
| ## | sex=F     | 21519 | 0.0629     | 0.7615 | 0.0631    | 0.0629   | 1.003       |
| ## | sex=M     | 27310 | 0.0724     | 0.7838 | 0.0722    | 0.0724   | 0.998       |
| ## | age=<50   | 9680  | 0.0584     | 0.7561 | 0.0616    | 0.0584   | 1.056       |
| ## | age=50-64 | 14083 | 0.0670     | 0.7845 | 0.0634    | 0.0670   | 0.945       |
| ## | age=65-79 | 15960 | 0.0690     | 0.7771 | 0.0680    | 0.0690   | 0.985       |
| ## | age=80+   | 9106  | 0.0790     | 0.7691 | 0.0829    | 0.0790   | 1.050       |

```
write.csv(tab, file.path(OUT, "16_tier5_subgroups.csv"), row.names=FALSE)
cat("\nTier 5 (fairness) complete. External eICU validation deferred (no data).\n")
```

```
##
```

```
## Tier 5 (fairness) complete. External eICU validation deferred (no data).
```